SEATTLEU
COLLEGE OF
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Age, Sex, Consumption and Cancer: A Support Vector Machine Approach

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INTRODUCTION

This study uses Support Vector Machines (SVMs) to predict cancer presence based on demographic and consumption factors.
The goal is to evaluate whether age, sex and diet variables (sweetened coffee, juice, sports drinks, alcohol use) can meaningfully predict cancer outcomes. Model performance is compared across linear, polynomial, and RBF kernels.

THEORETICAL BACKGROUND

SVMs classify data by finding an optimal decision boundary (hyperplane) that maximizes separation between classes. Kernel functions (linear, polynomial, RBF) allow modeling of nonlinear relationships.
Key parameters:
C: controls margin vs misclassification tradeoff
γ (gamma): controls influence of individual points
Performance evaluated using F1-score and ROC-AUC due to class imbalance.

METHODOLOGY

Dataset: 2022 NHIS (IPUMS Health Survey)
Target: Cancer (binary: 0/1)
Predictors: Demographics + lifestyle variables
Preprocessing:
  Median imputation for missing predictors
  Feature standardization
Modeling:
  Train/test split (70/30, stratified)
  SVM with linear, polynomial, and RBF kernels
  Hyperparameter tuning via cross-validation
Class imbalance handled using class weights

(I want to add more here… Should I add in optimal parameters for each kernel? Or would I add it to the result table?)

RESULTS

	Linear	Polynomial	Radial
Train accuracy	0.665	0.679	0.686
Test accuracy	0.671	0.679	0.660
Precision (Class 1)	0.25	0.25	0.24
Recall (Class 1)	0.81	0.79	0.81
F1-score (Class 1)	0.38	0.38	0.37
ROC-AUC	0.787	0.781	0.776

Figure 1. Performance result table for linear, polynomial and radial kernel

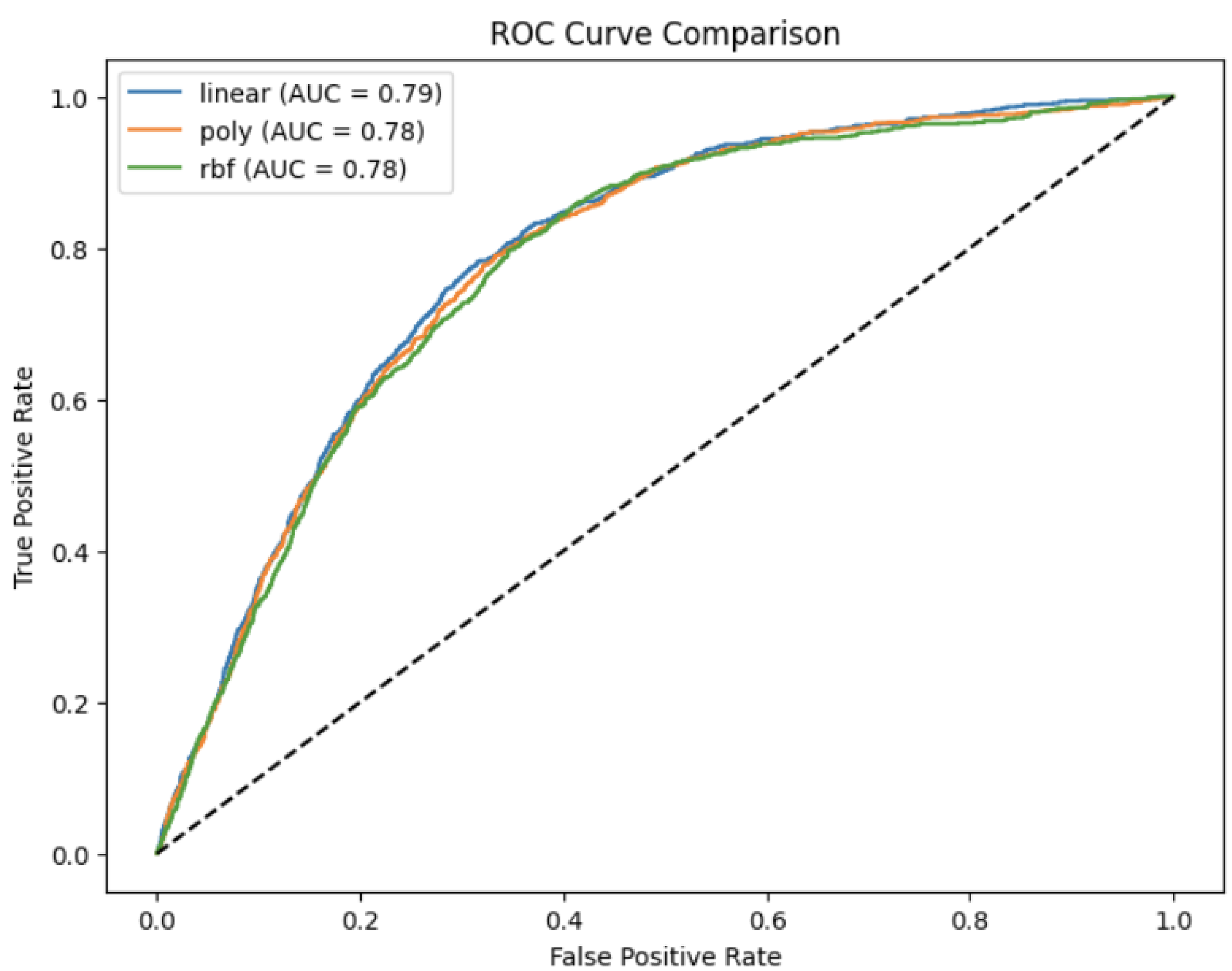

Figure 2. ROC curve comparison for linear, polynomial and radial kernel

KEY FINDINGS

- All models performed similarly (accuracy ≈ 0.66–0.68, ROC-AUC ≈ 0.78)
- Polynomial kernel achieved highest accuracy (0.679)
- Linear kernel achieved highest ROC-AUC (0.787)
- High recall for cancer cases (~0.80)
- Low precision (~0.25) → many false positives

Feature importance (Poly model):
- Age dominates prediction (0.85)
- Minor contributions from:
  - Tomato sauce (0.10)
  - Sex (0.05)
  - Alcohol & sports drinks (~0.02)

DISCUSSION

Model performance was consistent across kernels, suggesting limited nonlinear structure in the data.
Age is the primary driver of prediction, with behavioral variables contributing minimally.
High recall but low precision indicates the model identifies most cancer cases but over-predicts risk.
This tradeoff may be acceptable in screening contexts but limits practical deployment.

CONCLUSION

SVMs show moderate ability to predict cancer using survey data. Results highlight the dominant role of age in disease prediction.Behavioral variables alone provide limited predictive power.

Future work:
- Improve feature set
- Address class imbalance more aggressively
- Explore alternative models (e.g., ensemble methods)

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